Responsible Software - CheatSheet

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INTRO. **Responsible Software.** Software engineered in a responsible *(ethical)* way.
1. With a goal to do good. 2. While preventing avoidable negative impacts.
3. Taking people, other systems, social structures and our planet into account.
Active Responsibility. anticipating/preventing negative impacts before occurring.
Ethics. Principles/standards that guide human behavior in various contexts.
Metaethics. "what does it mean for an action to be right ?"
Normative. "how should people act in general (moral principles)?"
Applied. (Our approach) "how can i act ethically in a real-world situation?"

Technical decisions can have *ethical* implications.
Ethical Issue. any situation that may compromise respect of ethical value/principle.
Ethical Agency. capacity to take actions that align with ethical values and principles.
Ethical Sensitivty. ability to
- recognize ethical issues (awareness)
- identify their impact: who, what

Ethical Decision Making.
1. **Recognize**(knowledge+experience+habits)
2. **Evaluate**(cognitive+reason+affective) 3. **Act**.
Ethical Lenses.
1. **Desirability.** product corresponds to what people want.
2. **Viability.** product generates profits for a business.
3. **Feasibility.** product can be created with existing or new technology.
Fourth criterion (Isaac et al.).
Add **Ethicality.** product should be ethical.

Ethical Dilemmas. decision between multiple alternatives, all conflict with ethical values/principles. *None of the options override the other, wrong-wrong choice.*
Moral side.
Emotions should be used explicitely/intentionally for responsible design (Roeser). In response to moral issues or motivating immoral behavior.
Cognitive side (Normative Ethics).
- **Utilitarianism(Consequences).** choose the greatest good for greatest number
- **Deontology(Action).** do right according to universal principles
- **Virtue(Person).** act as a virtuous person
- **Care(Relationships).** mutual responsibility/care for each other
Trolley problem. thought experiment, **choice**: do nothing → runaway trolley kill people, or intervening → fewer people die. tests which moral principles justify action vs. inaction.
Ethical Decision Frameworks.
1. Identify the ethical issues
2. Get the facts
3. Evaluate alternative actions with *several ethical theories*
4. Choose an option and "test" it
5. Implement your decision and reflect on the outcome.
Example questions for brainstorming
- Who will use your product on a daily basis?
- Who is in geographical proximity with the users of your product?
- Who is in social proximity with the users of your product?
- Who is involved in the maintenance of your product?
- Who provides services that are needed to run your product?
- Who provides funding?
- Which communities are active in the application domain?

Stakeholder Individuals/groups/organizations/institutions/societies that might reasonably be affected by technology. **Direct Stakeholder.** *in contact* with the product
Indirect Stakeholder. affected through *consequences of the usage* of the software
Stakeholder analysis. Identifying every entity that can be affected by software positively/negatively.
1. Brainstorm to find all stakeholders possible using questions.
2. Use Direct vs Indirect Chart to find new stakeholders.
Additional Methodologies: *Ethix. add unintended stakeholders, Mitchell. add attributes to stakeholders, SAS. use different chart with scale for impact from product and influence on the product.*

Ethical Speculation. How to think, not what to think - anticipate potential impacts on stakeholders. Create a fictional story in two parts. (*done at the early design phase*)
I. Dark and Pessimistic
1. Imagine a character that would be negatively impacted by the software
2. Write a pitch that summarizes their story.
3. Find a title.
II. Positive outcome
1. Identify 1 or 2 main ethical issues represented in your story.
2. Describe the immediate and future consequences.
3. Imagine a happy ending for character.

SAFETY II. **Levels of analysis.** Macro (societies), Meso (groups/communities), Micro (individuals)
Internet Trafic. 99% submarine cables: 1.4M km, 4.7B users, 1.2 ZettaBytes (1.2 trillion GB).
Internet Benefits (+). Information access/dissemination, Communication, Social relations/organization, Identity/psychological development, Learning/cognitive development, Production/commerce.
Internet Harms (-). Privacy boundary violations, Social relationship harm, Community damage, Addiction, Cognitive impairment, Information overload, Knowledge/belief distortion, Democratic threats.
Tension. Same features enable both benefits/harms → *social connection enables social harm, information access causes overload, identity spaces blur privacy boundaries.*
From micro to macro.
Types of impacts: lots of individuals, Social structures/institutions, Different for different groups.
Areas: Markets/economy, Knowledge/worldviews, health/well-being, Politics/citizenship, Environment. ⚠**controversy** Complex measurements, causality hard to establish.

Disinformation. false info. with intent to deceive.
Misinformation. false info. without intent to deceive
Malinformation. true info. used with harmful intent.
fake news → false information, with or without intent.
Global risks. impact on electoral processes *(Brexit, U.S. elections)*, growth of distrust (government/media sources), risks of repression. *socio-technical phenomenon*
Cognitive effects. how we think.
Socio-affective effects. interaction with people and emotions.
System 1. fast/emotional/ activated first as quick response.
System 2. slow/logical/conscious analysis.
Illusory truth effect. repetition increases perceived truthfulness of claims by enhancing familiarity and intuitive/shortcut-based thinking.
Socio-affective factors
- **Source cues.** perceived credibility (powerful/attractive sources more likely to be believed).
- **False consensus.** overestimating how many people agree *(climate change denial: about 6% in reality but perceived as* ~ 50%*)*.
- **Emotions.** strong emotional content/receiver's emotional state increases susceptibility.
- **Identity resonance.** alignment with personal identity/values increases acceptance.
Continued influence effect: misinformation continues to influence beliefs even after correction is received and accepted as true.
Increased use of digital media shows greater exposure to misinformation, increased political polarization, decreased trust, and increased hate.
empirical results remain debated and sometimes exaggerated

Publication. publicly availabling content, *financial*/*political*/*coordinated strategic* incentives.
Human publishers. Influential figures/ordinary users publishing for visibility/revenue, often using *clickbait*/participating in organized campaigns *(coordinated troll operations)*.
Social bots. Automated accounts publishing at scale (**1–15%**), they generate *disproportionately large share* of misinformation.
Propagation. content spreading through *reposting*/*liking*/*algorithmic amplification*.
Human propagation. content reshared by real/fake accounts. engagement inflated by paid workers, while moderation reduces resharing *(account bans lowering repost rates)*.
Genuine resharing factors. Resharing driven by *inattention, emotional reactions*, and *socio-affective motives* such as signaling group membership or provoking reactions *(outrage)*.
Social bots in propagation. Automated accounts rapidly *repost* and *target influential or opposing users* to maximize visibility and conflict.
Recommender systems. *select, rank, display* content, optimize for *engagement* (*popularity bias*).

SAFETY I.
Security. Prevent the environment from affecting negatively the system. ⚠*threat* | *Vulnerability scanning, ethical hacking, security audit, code review*
Safety. Prevent the system from affecting negatively the environment. ⚠*hazard* | *Verification and testing, failure modes analysis, Redundancy*
Common challenge - human factor → consider software as **socio-technical system**
Internet, Autonomous vehicles, social media, human behavior and technical systems constanly influence each other.

Malfunction. software doesn't function/perform normally/as specified.
- Due to unintentional defect/fault/bug→critical consequences(⚠Life/Mission-critical systems)
Misuse/Abuse. attack, diversion, use of software for *nefarious purposes.*
Bad/threat actors. *malicious intent, authorized/unauthorized users*→direct/indirect impact.
Unintended use. use of software in a way not originally intended/not (officially) supported.
⊕ creates opportunities for innovation *use camera to scan documents*
⊖ *use llm to obtain factual information*
Intended use software used as intended. difficult to anticipate impact due to social ⚠scale/time

User-Generated Content (UGC). any content posted by users online *images/videos/texts*
- Published publicly and (usually) not produced by professional/institution *wikipedia*
⚠**Toxic content.** material harmful/offensive can have negative impact on individuals/communities
Types of impacts? Physical/Financial/Emotional/Relational.
Hate and Harassment.
Violence threats, Identity misrepresentation
Self-inflicted harm. Eating disorder promotion
Ideological harm.
Extremism/terrorism, Misinformation
Exploitation. Sexual abuse, Scams
Central dilemma. protecting users from toxic content while preserving free speech rights
To decide?
- State (responsible for fulfilling rights),
- Platforms (private actors with business interests),
- People (responsible for the content they publish.
Hate speech
- Discriminatory/pejorative of individual/group
- calls out real/perceived "identity factors" of individual/group
- conveyed through any form of (images/memes/symbols)
UGC platforms provide channels for hate speech. Easily produced, shared, anonymous, global/diverse audience, realtime, amplified by content recommendation algorithms
Free speech. right to express opinions and to seek, receive, share information/ideas, in any format offline/online, subject to limitations that respect the rights of others and prohibit incitement to discrimination or violence, pillar of democracy.

Predicted \ **Actual**	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)

Content moderation:
False negative: toxic content that escapes moderation.
False positive: wrongful censorship, restricting free speech.

TP. Model predicts positive and is correct
Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
Positive Predictive Value (PPV) = $\frac{TP}{TP+FP}$
Negative Predictive Value (NPV) = $\frac{TN}{TN+FN}$
False Discovery Rate (FDR) = 1 − PPV = $\frac{FP}{TP+FP}$
False Omission Rate (FOR) = 1 − NPV = $\frac{FN}{TN+FN}$
False Positive Rate (FPR) = $\frac{FP}{FP+TN}$
False Negative Rate (FNR) = $\frac{FN}{FN+TP}$

Bad Actors Modelling Minimize vulnerabilities via risk assessment and mitigation.
When to Use: Design phase, prior to updates, audits, incident responses, and improvements.
What is a Bad Actor? A person/group exploiting a system for personal goals.
Motivations (might overlap):
- Money: Data theft, ransomware, sabotage by competitors.
- Politics: Misinformation spreading, strategic sabotage, infrastructure attacks.
- Entertainment: Attacks on random systems, misinformation spreading.
- Self-interest: Harassing messages, communication monitoring, location tracking.
- Ideas: Discrediting groups, data leaks, website defacement.
Steps: 1. Identify possible malicious users and their motivations. 2. Assess potential consequences for the system and users. 3. Develop measures to prevent these negative outcomes.

Harm Modelling anticipate/address potential harms users/stakeholders may encounter.
When to Use: At every stage of project, for every feature, until harm potentials are satisfactory.
Evaluate: 1. The category of use. 2. the types of harms. 3. the magnitude of harm

Categories of Use:	**Types of Harm:**	**Magnitude of Harm:**
Intended Use: Harm from correct usage.	*Humans:* Physical, emotional (or psychological) injury.	*Severity:* Impact on individual/group well-being.
Unintended Use: Harm from unexpected usage.	*Allocation of Resources:* Opportunity, economic losses.	*Scale:* How broadly the harm affects populations.
Misuse/Abuse: Harm from malicious usage.	*Human Rights:* Dignity, liberty, privacy losses; environmental impact.	*Probability:* Likelihood of harm occurring.
Malfunction: Harm from system failures.	*Social Systems:* Manipulation, social detriment.	*Frequency:* How often the harm might occur.